Breaks: 2 Coffee break — 10:30 am IST, 3:00 pm IST — 15 mins
1 lunch break - 1 pm IST — 1 hr
Day 1 = 4 hours → 1 break @ 10:30 am IST (15 mins)

Yes = +1, No = +2

Models - ? → Github Copilot ?

FM / LLM

Models → ML Models ?

FM / LLM → Traditional ML Models → No → 1

Providers
- Anthropic → claude
- Meta → Llama
- Mistral → Mistral
- Google → Gemini
- OpenAI → GPT

Data set → Train → ML Model → understand patterns → predict the output
Captured → Algo

AI → ML → Deep Learning → Gen AI
↳ creating something New
Powered by FM / LLM
Content (Text / Code), Image, Video, Audio
↳ Fresh.

Foundation Model (FM) / LLM
↳ Pre-trained Models which are trained on [Internet-scale of data]

Prompt: Text, Image, video, etc. → Input → Model (FM) → output → Text, Image, Video

Dataset [Text, Image, video, audio, HTML, JSON, Text, etc.] → Raw Data → Model → output
↑ General Domain

(Raw) Multi Modal Data → Transformer Architecture

FM → LLM (Large Language Model) → Input as Text, Image, Video, Audio → output as Text ONLY
① Subset of FM
② Text generation, code generation, Text summarization, Conversation, Q&A, RAG, etc.

Core Components of Transformer
① Input Embedding = Input → ① Tokens → ② numerical representation (vectors)
[Translate | from | Gujarat | in | French]
→ 1 | 1 | 1 | 1 | 1 | 1 | 1
Model (Claude) → (all of tokens) → Input tokens + output Tokens
② Encode → Decode → output

Providers
→ Anthropic → [Claude] → Chatty Model
→ OpenAI → GPT

Microsoft Learn AC ?
↳ Achieve Achievement Code.

Billing (Github Copilot)
[Requests] → Any interaction → Asking Copilot to do something
→ Generate the code
→ Answering the Qs.
→ Help you things the outcomes.

Premium Req. (3.33 x) → Premium features
① Copilot chat
② Copilot CLI
③ Copilot Code reviews
④ Copilot coding agent

Copilot Free → 2000 inline suggestions
50 premium reqs

Paid plan → [Unlimited inline suggestions
Unlimited chat interactions (GPT-5 mini, GPT 4.1, GPT-4o)
Monthly allowance of premium reqs.]

Unused Reqs for the previous month → X carry over

AI Models Comparison
① General-purpose coding & writing (Fast, Accurate code completion & explanations) → GPT 4.1, GPT-5 mini
② Deep Reasoning & Debugging → GPT 5.1, GPT 5.1-Codex
③ Agentic S/W Development → GPT 5.1 Codex Max
④ Fast help with simple & Repetitive Tasks → Claude Haiku
⑤ Deep Reasoning & Debugging → Claude Opus
⑥ General purpose Coding & Agent Tasks → Claude Sonnet
⑦ Working with Visuals (diagrams, screenshots) → GPT 5 mini, Claude Sonnet 4.6, Gemini 3 Pro

Interact with Copilot

① Inline suggestions
① To accept a suggestion → Tab key / right arrow key
② To reject the suggestion → keep typing / Esc key
+
Invoke Inline chat ? → ctrl + I

② Slash Commands → Shortcuts
/explain

Basic web app scaffold with Github Copilot